LLM-Assisted Risk-Adaptive Neurosymbolic Intent Planning for Optical Networks

A Pre-Deployment Decision Mechanism with Joint Semantic and QoT Assessment

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OUTLINE

[01] Bottleneck

The Optical Intent Planning Bottleneck

Physical-layer constraints, token saturation, and hallucinated routing in optical backbones

[02] Architecture

Neurosymbolic Intent Planning Pipeline

Decoupling probabilistic reasoning ($\mathcal{I}_{NL} \rightarrow \mathcal{S}_{PDDL}$) from deterministic solvers and physics tools

[03] Risk Gates

Pre-Deployment Risk Gates: Semantic & Physical Validation

Sequential fail-fast decision via Layer 1/2 semantic gate (U_{sem}) and GN-model QoT gate (QoT_{valid})

[04] Evaluation

Experimental Testbed Validation on 17-Node Optical Topology

Benchmarking safety ($UAR = 100\%$), human intervention reduction, and orchestration latency against baselines

[05] Outlook

Key Takeaways, System Guarantees & Future Directions

Summary of thesis contributions, operational guarantees, and extension to joint compute scheduling

Motivation: The Vision of Intent-Based Optical Networks

⚠️ Operational Shift: Manual Bottleneck

- Optical backbones carry terabits of core traffic across ROADM networks
- Traditional workflow: manual CLI scripts and complex RESTConf payloads
- Human configuration delays lightpath provisioning by hours or days
- High cognitive load and misconfiguration risk across multi-vendor links

🎯 Goal: Transition to autonomous Intent-Based Networking (IBN)

⚡ Operational Promise: Autonomous Vision

- High-level abstraction: specify what is needed, not how to configure it
- Realistic carrier intent: 'Establish a 400G lightpath between Milan and Rome avoiding link L2'
- Autonomous translation into verified, collision-free physical lightpaths
- Rapid sub-second provisioning reducing operational delays by orders of magnitude

⚠️ Critical Challenge: Optical networks do not tolerate probabilistic errors

Operational Flow: [Operator NL Intent] ➔ [AI Intent Orchestrator] ➔ [Zero-Error Physical Lightpath]

Illustrative Failure: Why Standard LLMs Break Optical Backbones

⚠️ 1. Token Budget Saturation	Telemetry dumps trigger attention degradation, dropping critical route exclusions
🚫 2. Hallucinated Physics	Probabilistic predictors lack wave propagation engines, violating non-linear $GSNR$ margins
🔄 3. Semantic Drift	Unconstrained multi-turn conversational loops mutate or drop initial boundary constraints
⏱️ 4. Reactive Deployment Latency	Trial-and-error configuration risks live outages and introduces high control-plane recovery latency
👥 5. Suboptimal HITL Friction	Binary all-or-nothing review causes operator fatigue or outages; models fail to fail-early

Empirical Risk: Unconstrained LLMs allow up to 34% unfeasible deployments, semantic drift loops, and critical control-plane latency

Problem Statement: Given, Decide, Objective & Constraints

[1] Given (System Inputs)

- Unstructured operator intent: $\mathcal{I}_{NL}$
- Active optical topology graph: $G(V, E)$ via RESTConf
- Physical parameters: span length L, attenuation α, gain G_m
- Feasibility threshold: $GSNR_{th} = SNR_{min} + Margin$

[2] Decide (Variables & Actions)

- Formal symbolic specification: $\mathcal{S}_{PDDL}$ compiled from intent
- Optimal physical lightpath route: $\pi^* \in \mathcal{K}_{path}$ from candidate paths
- Pre-deployment control action: $a \in \{approve, clarify, replan\}$
- Provisioning routing configuration: c^* dispatched to controller

[3] Objective (Optimization)

- Minimize operational friction: $min = \alpha \cdot N_{hitl} + \beta \cdot T_{tokens}$
- Cut operator cognitive fatigue: $min\ N_{hitl}$ (minimize human interruptions)
- Bound compute cost and latency: $min\ T_{tokens}$ (minimize prompt tokens)
- Hard pre-deployment safety guarantee: $\mathcal{D}(U_{sem}, QoT_{valid}) = approve$

[4] Constraints: Resource Limits vs. Physical & Semantic Boundaries

Resource Constraints (System & Solver Limits):

- Localized prompt context window bound: $T_{prompt} \leq T_{max} \ll T_{full}$ (via G_{sub})
- Strict end-to-end execution latency budget: $t_{exec} \leq t_{max\_budget}$
- Bounded path search complexity: $K - SP$ with $K \in [3, 5]$

Boundary Constraints (Physical & Semantic Feasibility):

- Zero semantic drift tolerance bound: $U_{sem} \leq \tau_{sem}$
- Deterministic optical QoT feasibility: $GSNR(\pi^*) \geq GSNR_{th} \wedge P_{rx}(\pi^*) \geq P_{rx,min}$
- Pre-deployment physical validity state: $QoT_{valid} \in \{0, 1\}$

Proposed Solution: Neurosymbolic Decoupling

🧠 Neural Subsystem (Semantic Domain)

- Translates unstructured natural language $\mathcal{I}_{NL}$ into formal PDDL $\mathcal{S}_{PDDL}$
- Zero routing arithmetic or physical SNR calculations performed by LLM
- Reverse prompting reconstructs intent $\mathcal{I}_{recon}$ for validation
- Interpretable intermediate representation prevents hallucinated configurations

📐 Symbolic Subsystem (Optical Domain)

- Subtopology extraction via scoped Mock GraphRAG ($k - hop$ neighborhood)
- Deterministic routing via Yen's $K - SP$ constrained graph algorithm
- Physical QoT validation via pure Python analytical GN-model engine
- Evaluates $GSNR \geq GSNR_{th}$ and $P_{rx} \geq P_{rx,min}$ with zero hallucination

🎯 Core Thesis Contributions (Architectural Novelties)

1. Neurosymbolic Decoupling & Scoped GraphRAG

- Strict separation: probabilistic semantic translation vs deterministic physics
- Subtopology scoping reduces prompt tokens by >75%, eliminating attention loss

2. Dual-Layer Semantic Uncertainty Gate (U_{sem})

- Layer 1 syntax check ($v_{struct} \in \{0,1\}$) + Layer 2 semantic discrepancy (d_{sem})
- Pauses via LangGraph interrupt() when $U_{sem} > \tau_{sem}$ to clarify ambiguity

3. Risk-Adaptive Decision Gate (RADG) & QoT

- Piecewise decision $D(U_{sem}, QoT_{valid})$: clarify, replan, or auto-approve
- Guarantees $UAR = 100\%$ physical safety with <5 ms calculation latency

End-to-End System Architecture & Pipeline Flow

Phase 1
Optical RAG
Enrich $\mathcal{I}_{NL}$ with ITU-T grid & transponders

Phase 2
PDDL Parser
Translate intent into formal PDDL $\mathcal{S}_{PDDL}$ AST

Phase 3
Semantic Gate
[RISK GATE]
Evaluate CFG v_{struct} & Reverse Prompting d_{sem}

Phase 4
Symbolic Solver
Scoped GraphRAG & Yen's $K-SP$ routing

Phase 5
QoT Physics
Deterministic GN-model $GSNR$ calculation

Phase 6
Risk Gate (RADG)
[RISK GATE]
Piecewise risk gate $D(U_{sem}, QoT)$

Phase 7
Plan Synthesizer
Auditable report & deployment commands

⏸ Phase 3b: Clarify Loop (LangGraph interrupt())
If $U_{sem} > \tau_{sem}$: Pauses execution and prompts operator for intent disambiguation

↺ Phase 6: Suggest Replan Loop (Physics Unfeasible)
If $QoT_{valid} = 0$: Suggests operator to relax constraints (lower baud rate, alternate link)

Overcoming Token Saturation: Scoped Optical GraphRAG

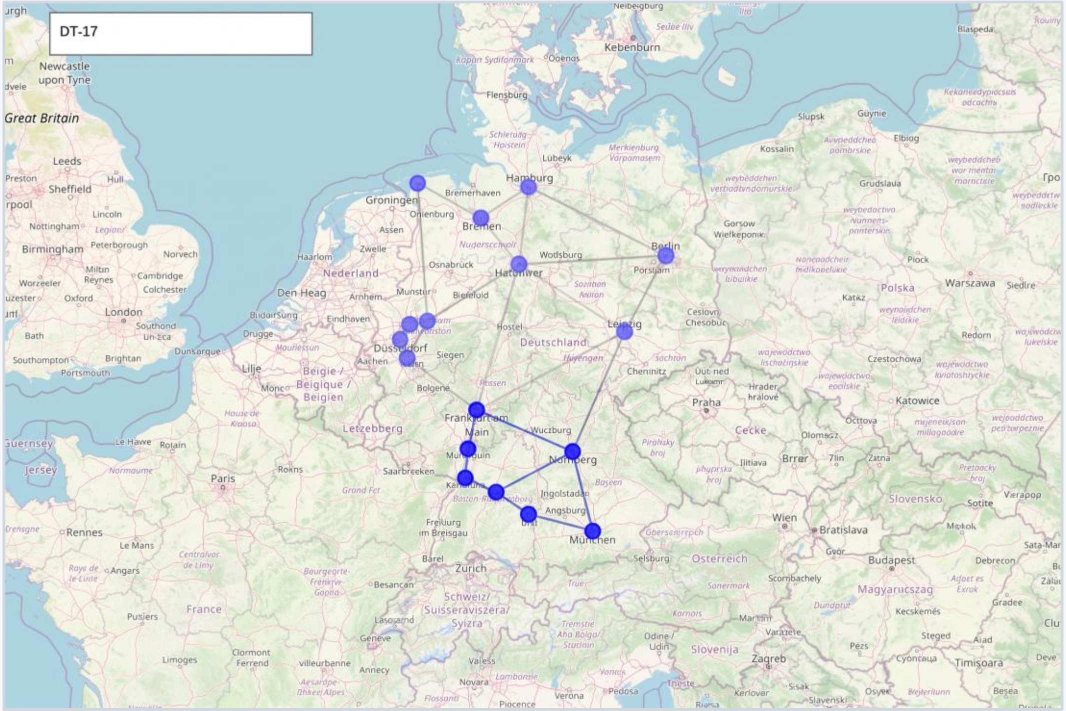
🌐 Deterministic k-hop Subtopology Scoping

- Full topology dumps overwhelm LLM context windows
- Raw JSON contains excessive telemetry: ROADM ports, EDFAs, fibers
- Mock GraphRAG extracts localized subtopology $G_{sub} \subseteq G$
- Bounding $k - hop$ neighborhood between candidate endpoints
- Quantitative Impact: Over 75% reduction in prompt token payload
- Sub-millisecond graph extraction: $\mathcal{O}(|V| + |E|)$ executed in pure Python
- Guarantees sharp LLM attention focus on active optical constraints

Mathematical Scoping Bound:

$$G_{sub} = (V_{sub}, E_{sub}) \subseteq G$$
$$T_{prompt}(G_{sub}) \ll T_{prompt}(G)$$

🗺️ 17-Node German Backbone: Subnetwork Scoping



⚡ Frankfurt ➔ Munich demand: Northern nodes pruned from LLM prompt

Overcoming Semantic Drift: Reverse Prompting & HITL

🧠 Two-Layer Semantic Uncertainty (U_{sem})

- Layer 1 (Structural): CFG regex AST check ($v_{struct} \in \{0,1\}$) catches syntax errors
- Layer 2 (Semantic): Reverse Prompting reconstructs NL $\mathcal{I}_{recon}$ directly from PDDL
- Independent LLM judge measures semantic discrepancy $d_{sem} \in [0,1]$

Uncertainty Formulation:

$$U_{sem} = \begin{cases} 1 & if\ v_{struct} = 0 \\ d_{sem} & if\ v_{struct} = 1 \end{cases}$$

⏸️ Fail-Fast HITL Clarification Loop

- Evaluated BEFORE invoking graph solvers or physical tools
- If $U_{sem} > \tau_{sem}$: Pipeline pauses via LangGraph interrupt()
- Prompts operator to clarify ambiguous parameters or missing nodes
- Eliminates infinite trial-and-error conversational loops
- Guarantees downstream tools receive mathematically verified intent
- If operator approves valid syntax ($v_{struct} = 1$): Bypasses parsing straight to solver

Fail-Fast Guarantee: Zero computational waste on physics simulation when intent is ambiguous

Pre-Deployment Risk Gate: The RADG Decision Function

Piecewise Decision Formulation $D(U_{sem}, QoT_{valid})$:

$$D(U_{sem}, QoT_{valid}) = \begin{cases} clarify & if\ U_{sem} > \tau_{sem} \\ replan & if\ U_{sem} \leq \tau_{sem} \wedge QoT_{valid} = 0 \\ approve & if\ U_{sem} \leq \tau_{sem} \wedge QoT_{valid} = 1 \end{cases}$$

❓ Clarify Intent

Condition: $U_{sem} > \tau_{sem}$

- Trigger: Semantic uncertainty exceeds threshold
- Action: Pause pipeline via interrupt()
- Prompts operator to provide missing data
- Prevents unverified intent from reaching physics

↺ Suggest Replan

Condition: $U_{sem} \leq \tau_{sem} \wedge QoT_{valid} = 0$

- Trigger: Valid semantics, but physics infeasible
- Action: Physics failed; notifies operator
- Suggests relaxing constraints (e.g. lower baud rate)
- Loops back to PDDL parsing with feedback

✓ Auto-Approve

Condition: $U_{sem} \leq \tau_{sem} \wedge QoT_{valid} = 1$

- Trigger: Clear semantics and feasible $GSNR$
- Action: Autonomous zero-touch provisioning
- Generates auditable planning report
- Zero unverified configurations reach controller

Deterministic Physical Layer: GN-Model QoT Validation

🔬 Analytical Gaussian Noise (GN) Engine

- Pure Python implementation: zero LLM arithmetic

Accumulated ASE Noise Power per Span:

$$P_{ASE} = (G - 1) \cdot h \cdot \nu \cdot F \cdot B_{ref}$$

Nonlinear Interference (NLI) Noise Power:

$$P_{NLI} \approx \eta \cdot P_{ch}{}^{3}$$

- Accounts for fiber Kerr nonlinearity, dispersion, and EDFA noise

📐 Optical Feasibility & GSNR Criteria

Generalized Signal-to-Noise Ratio (GSNR):

$$GSNR = \frac{P_{ch}}{P_{ASE} + P_{NLI}} \geq GSNR_{th}$$

Receiver Power Sensitivity Budget:

$$P_{rx} = P_{launch} - A_{total} + G_{total} \geq P_{rx,min}$$

- Deterministic feasibility: $GSNR \geq GSNR_{th} \wedge P_{rx} \geq P_{rx,min}$

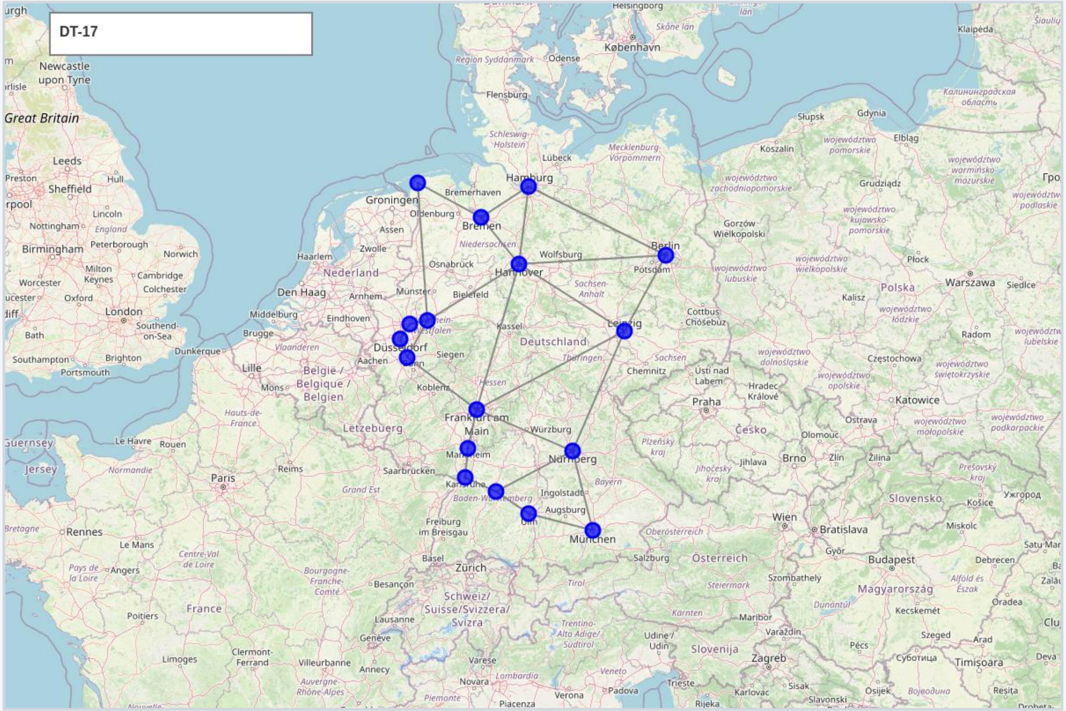
⚡ **Execution Speed:** $T_{phys} < 5\ ms$ **per candidate route (100% deterministic reproducibility)**

Experimental Setup & Testbed Environment

🌐 17-Node German Core Network Benchmark

- Realistic telecom benchmark: $|V| = 17, |E| = 26$ bidirectional fiber links
- Standard Single-Mode Fiber (SMF-28): $\alpha = 0.2\, dB/km$
- Dispersion parameter $D = 16.7\, ps/(nm \cdot km)$, $\gamma = 1.2\, W^{-1}km^{-1}$
- Amplified spans: dual-stage EDFAs with noise figure $NF = 5.5\, dB$
- Dynamic span lengths ranging from $L \in [45, 350]\, km$
- Orchestrator: LangGraph StateGraph with memory persistence
- Telemetry: RESTConf and Mock SDON testbed client adapters

🗺️ Nobel-Germany 17-Node Optical Core Backbone



⚡ 17 ROADMs, 26 amplified fiber links, RESTConf telemetry

Evaluation Framework & Benchmark Scenarios

⚖️ Architectural Baselines

- Baseline A (LLM-Only): Direct prompt-to-configuration generation with heuristic retry
- Baseline B (Static Rule-Based): Strict regex parser with always-on human review
- Proposed (Neurosymbolic RADG): Decoupled translation + sequential risk gates
- Evaluates LLM reasoning limits vs deterministic safety

🧪 100 Test Demands (4 Classes)

- Nominal Intents [40]: Unambiguous requests with feasible physics
- Ambiguous Intents [20]: Under-specified constraints triggering $U_{sem} > \tau_{sem}$
- Infeasible Intents [25]: High modulation over long spans ($QoT_{valid} = 0$)
- Adversarial Prompts [15]: Hallucinated nodes ($v_{struct} = 0$) & syntax violations

📊 Evaluation Metrics

- Pre-Deployment Blocking Accuracy: $UAR = 100\%$ guarantee
- Human Intervention Rate (HIC): % demands requiring operator clarification
- End-to-End Latency (T_{E2E}): NL intent to planning report turnaround
- Token Consumption (T_{prompt}): Overhead across scoped vs full topologies

Key Findings & Pre-Deployment Guarantees

100%

Pre-Deployment Safety Guarantee

$UAR = 100\%$: Zero unfeasible lightpaths reach controller (vs 34% in baseline)

> 70%

Reduction in Operator Fatigue

Autonomous approval when $U_{sem} \leq \tau_{sem}$ (operator engaged only on high risk)

< 5 ms

Deterministic Physics Latency

GN model executes in $T_{phys} < 5\ ms$ ($T_{solver} < 10\ ms$, sub-second total)

Empirical Benchmark: GSNR Distribution & Blocking Rate vs Baselines

(Target: 16:9 Matplotlib / Chapter 4 Benchmark Plot — Replace in PowerPoint)

Conclusions & Main Takeaways

🧠 Neurosymbolic Decoupling is Essential

- Probabilistic LLMs must never calculate physical impairments or route lightpaths
- Natural language reasoning $\mathcal{I}_{NL} \rightarrow \mathcal{S}_{PDDL}$ bridges to formal planning
- Guarantees formal soundness without constraining operator expressiveness

🔒 Sequential Risk Gates Prevent Failure

- Early semantic evaluation ($U_{sem} \leq \tau_{sem}$) eliminates drift before physics computation
- Deterministic GN-model verification ($QoT_{valid} = 1$) ensures 100% physical feasibility
- Zero unverified configurations ever reach the optical controller

⚡ Selective HITL Optimizes Efficiency

- Engaging operators proportionally to risk eliminates fatigue while maintaining safety
- Over 70% reduction in human intervention compared to always-on review
- Sub-second planning latency enables rapid dynamic lightpath turn-up

⚙️ Production-Ready Engineering

- Fully implemented LangGraph state machine with memory checkpoints
- Benchmarked on realistic $|V| = 17, |E| = 26$ German topology and RESTConf testbed
- Establishes a reproducible foundation for autonomous optical networking

Future Outlook & Acknowledgments

🖥️ Joint Compute and Optical Scheduling

Extending PDDL domain to co-schedule distributed GPU cluster workloads alongside optical lightpaths

🌈 Multi-Band & Dynamic Optical Channels

Expanding analytical GN-models to incorporate C+L multi-band SRS inter-channel Raman crosstalk

🔄 Autonomous Online Self-Healing

Ingesting real-time testbed telemetry for closed-loop intent re-planning upon physical fiber degradation

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Thank You for Your Attention!

Questions & Discussion